DEV PATEL

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EDUCATION

Saint Joseph's University — College of Arts and Sciences *Philadelphia, PA*
Bachelor of Science in Computer Science Expected May 2028

- **GPA:** 3.83 / 4.0 **Awards:** Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026), Honors Program, Presidential Scholarship
- **Relevant Coursework:** Data Structures, Design & Analysis of Algorithms, Computer Architecture, Principles of Programming Languages, Software Engineering, Discrete Structures, Penetration Testing, Calculus I–II

TECHNICAL SKILLS

- **Languages:** C, C++, MATLAB, Python, Java, C#, Bash, JSON
- **Tools & Platforms:** Git, GitLab, SVN, CMake, GoogleTest, Google Benchmark, Polyspace, JFrog, Polarion, Jira, VS Code, CAN tooling, PC/HIL simulation, UiPath, JD Edwards EnterpriseOne, Microsoft 365 Tools
- **Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Embedded Systems, CI/CD, Agile/Scrum, Version Control & Migration, Software Testing & Validation, Performance Optimization, Debugging
- **AI Tools:** Experience working with multiple AI models, including Claude and GPT

PROFESSIONAL EXPERIENCE

CNH Industrial *New Holland, PA*
Embedded Software Engineer Intern May 2026 – Aug 2026

- Contributed embedded **C/C++** software to the release of **PV25** for the cotton harvester product line, developing, testing, and maintaining features for the machine's onboard control systems.
- **Led the company-wide migration from SVN to GitLab**, planning and executing the transition of repositories and CI/CD pipelines and building Git/Bash tooling and documentation adopted by software teams across multiple international sites.
- Developed and tested embedded control components in **MATLAB and C**, validating changes on a **PC-based hardware simulator** and with **CAN bus tooling** before deployment to physical machines.
- Configured and troubleshot control systems, sub-systems, and components on the cotton harvester, resolving system-level issues and gaining hands-on systems-engineering experience.
- Collaborated daily with distributed engineering teams across **Belgium, Brazil, Italy, India, and the U.S.**, participating in Agile ceremonies, peer code reviews, and **Polyspace static analysis** to detect defects early.
- Managed requirements, traceability, and change control in **Polarion**, using **JFrog** and GitLab CI/CD in accordance with ISO-compliant software development processes.
- Worked in **Linux and Ubuntu** environments, and applied **Claude, GPT, and Gemini** AI models in daily tasks and software development projects.

Morgan Corporation *Morgantown, PA*
Information Technology Intern Aug 2023 – Mar 2024

- Automated repetitive business processes by building **UiPath robots with C# integration**, shortening multi-step manual workflows.
- Developed and debugged **JD Edwards EnterpriseOne** pages within the company's ERP system.
- Organized the JD Edwards redesign and ERP implementation plan using Microsoft Excel.
- Responded to security incidents, provided technical support, and provisioned security access for employee onboarding and temporary project members.

PROJECTS

Limit Order Book & Matching Engine *github.com/patelgroup33/limit-order-book*
C++ · CMake · GoogleTest · Google Benchmark · GitHub Actions Summer 2026

- Built a production-inspired **matching engine with price–time (FIFO) priority** supporting limit and market orders, partial fills, cancellation, and modification with correct time-priority semantics.
- Engineered a **zero-allocation hot path** using intrusive linked lists and an object pool; benchmarked an initial std::map design, then replaced it with a **direct-indexed price ladder**, flattening add/cancel from **O(log P) to O(1)** and cutting per-fill match latency **~2.2×**.
- Designed a **strongly-typed domain** (Price/Quantity/OrderId) that turns argument-swap bugs into compile errors at zero runtime cost, and made matching **deterministic and replayable** via monotonic sequencing.
- Validated with **54 GoogleTest unit tests** and Google Benchmark microbenchmarks; automated Linux/macOS CI with **CMake and GitHub Actions**, and built a dependency-free web visualizer (L2 depth ladder, market depth, time-&-sales) deployed via GitHub Pages.
- Integrated **Claude** to handle portions of the coding workload while owning backend logic and system architecture.

Backtesting Engine *github.com/patelgroup33/limit-order-book/tree/main/libs/backtest*
C++ · CMake · GoogleTest · Google Benchmark Summer 2026

- Built a modular, **event-driven backtesting framework** that replays historical market data through pluggable strategies with no **look-ahead bias** — the same strategy code runs in backtest and live.
- Simulated realistic fills with **slippage, commission, partial fills, and latency**, and tracked positions, cash, and **realized/unrealized PnL** with average-cost accounting and a full equity curve.
- Computed **performance analytics** (Sharpe, Sortino, max drawdown, CAGR, profit factor) and gated orders through a **risk manager** with position/exposure limits and a daily-loss kill-switch.
- Benchmarked **10M-bar backtests at ~50–70M bars/sec** (O(N)); validated with **100+ unit tests**; built a browser visualizer replaying price, trades, and the equity curve.
- Integrated **Claude** to handle portions of the coding workload while owning backend logic and system architecture.